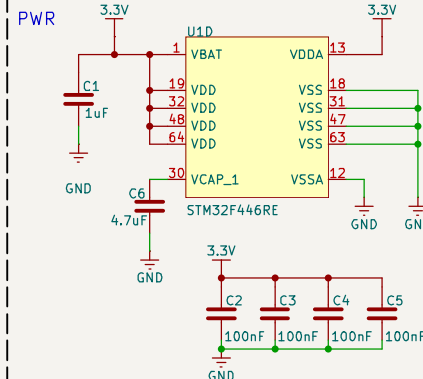
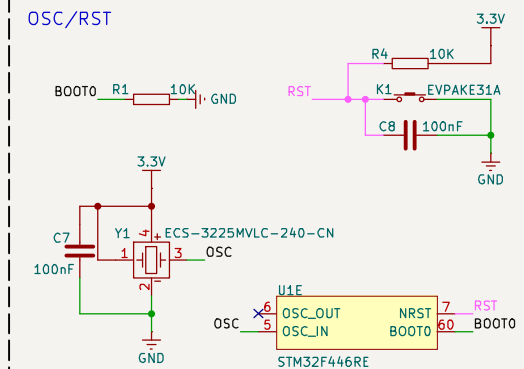


GPIO



OSC/RST



Pinout diagram for the 40-pin J1 connector of the 52991-0308 module. The diagram shows two rows of pins. The left row (pins 1-16) is connected to various CAN and VCP signals: CAN.TX (pin 1, red), CAN.RX (pin 3, red), VOLT.V (pin 5, red), VOLT.W (pin 7, red), VCP.RX (pin 9, green), VCP.TX (pin 11, green), RST (pin 13, cyan), SWO (pin 15, cyan), SWCLK (pin 17, cyan), SWDIO (pin 19, cyan), USB.DP (pin 21, magenta), USB.DN (pin 23, magenta), and VOLT.SRC (pin 25, yellow). The right row (pins 17-30) is connected to current and voltage sense signals: CURR.U (pin 17, green), CURR.V (pin 19, green), VOLT.U (pin 21, green), TMC.DIAG (pin 23, magenta), TMC.UH (pin 25, magenta), TMC.VL (pin 27, magenta), TMC.WL (pin 29, magenta), and TMC.WL (pin 30, magenta). A 3.3V supply is connected to pin 2, and a 5V supply is connected to pin 18. A ground connection is shown for pin 20.

Signal	Pin	Signal	Pin
CAN.TX	1	CURR.U	17
CAN.RX	3	CURR.V	19
VOLT.V	5	VOLT.U	21
VOLT.W	7	TMC.DIAG	23
VCP.RX	9	TMC.UH	25
VCP.TX	11	TMC.VL	27
RST	13	TMC.WL	29
SWO	15	TMC.WL	30
SWCLK	17		
SWDIO	19		
USB.DP	21		
USB.DN	23		
VOLT.SRC	25		

52991-0308

[illegible]

ENC.CS
ENC.CLK
ENC.POCI
ENC.PICO

U20 AS5048A

C5
CLK
MISO
MOSI

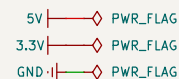
VDD3V
VDD5V
GND

3.3V

C27
C28

100nF
10uF

GND



Title: BLDCM MCU

Rev: v1.0
Id: 1/1